

# INSTITUTE OF ACTUARIES OF INDIA

## EXAMINATIONS

21<sup>st</sup> March 2017

**Subject SA6 – Investment**

**Time allowed: Three Hours (14.45\* – 18.00 Hours)**

**Total Marks: 100**

### INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *\* You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

#### AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** The Central Bank of your country is considering “demonetizing” its high value currency notes. These account for about 80% of the currency in circulation. These notes would cease to be legal tender from the day the demonetization drive is likely to get announced by the Central Bank. Also note that more than 90% of transactions in the country are carried out using cash.

The Government will then give an option to exchange old notes for new notes – to people who possess cash in hand from sources of income upon which taxes have already been paid. For others, they have to pay a penal rate of tax, a flat 60%, and then exchange the balance for new notes.

Further, the exchange of notes is restricted to Rs.10,000 per week for the next 4 months. Thereafter these limits would be gradually increased.

You are an investment actuary working in an investment consultancy that advises Foreign institutional investors regarding their investment in India. You have been asked to share your views on the consultancy’s monthly newsletter to the clients on the following matters :

- i)** Assuming that the Central Bank opts to go ahead with the demonetization program, elaborate on the likely impact of the same in the near future. Your response should cover impact on inflation, interest rates, taxation, and economic growth. (12)
- ii)** Elaborate on the possible effects of demonetization on the following asset classes stock prices on the following asset classes/sectors:
- Real Estate
  - Government Securities and Corporate Bonds
  - Infrastructure
  - Gold and Jewellery
  - Banking
  - NBFCs and Housing Finance Companies
  - Oil and Gas
  - Automobiles and Consumer Durables
  - Pharmaceuticals
  - Listed Online Mobile Payments Company. (20)
- iii)** Based on your comments above, elaborate on investment strategies you would suggest to your institutional investors in order to hedge against losses or to profit from the inherent uncertainty. (6)
- iv)** A senior consultant in your organization feels that demonetization is positive news for the mutual fund industry. What could be the possible reasons for the consultant to expect a positive impact for mutual funds (2)

**[40]**

- Q. 2)** You work for an investment-consulting firm. Your firm has been engaged by the trustees of a large Pension scheme in India. The assets under the Pension scheme are self managed and are largely invested in Government securities and bonds instead of equities.

Even though long duration bonds have been chosen for investments, the asset duration is less than the liabilities. It is also expected that interest rates will continue to remain low and may show further decline in the years to come. Hence the trustees have emphasized on the need of active asset liability matching. However the fund has never used derivatives.

You are a qualified actuary holding a senior role in your firm and have good understanding of the investment market in India and asset liability management issues. On behalf of your firm you are the lead consultant to the trustees of the Pension fund.

- i)** The trustees have asked you to explain in detail what “Liability Driven Investment” implies and how it can be used as an investment strategy to manage the following risks:
  - a)** Interest rate risk
  - b)** Inflation risk
  - c)** Liquidity risk arising due to varying patterns or shapes of the liabilities i.e. the risk of not being able to support the actual cash payouts when they fall due (12)
- ii)** Please explain why the pension company would have adopted a matched bond based investment strategy over these years? (5)
- iii)** List different ways of reducing duration mismatch risks (3)
- iv)** The Pension fund has never invested in derivatives. However the management feels that it can be effective to reduce the risks associated with falling interest rates. Hence they want to evaluate interest rate swaps, swaptions and interest rate futures and they seek your advice on the effectiveness of these instruments. (8)
- v)** The trustees also want advice on investments in ETFs. However the investments will still largely be in debt assets. Please explain what is an ETF and discuss the advantages and disadvantages of gaining exposure to debt assets like bonds through ETFs (5)
- vi)** Investment strategies or policies are also driven by other requirements or restrictions on the scheme. For example the minimum funding requirement will imply that the sponsoring employer needs to pay-in extra contributions into the scheme in case there is a fall in the funding level, below a certain trigger point. Please explain what are the key drivers for Investment Policy of a Pension scheme (3)
- vii)** Please outline the process of using Asset-Liability Modeling to arrive at an appropriate investment strategy for the pension scheme. (5)
- viii)** Please state the assumptions to be made in the Asset Liability Modeling exercise of a Pension scheme. (4)
- ix)** Other than an appropriate investment strategy, what additional management insights, can be gathered from the Asset Liability Modeling exercise. (3)

- x) It might be argued that the traditional actuarial valuation considers interaction of assets and liabilities. Why then would you require to conduct detailed asset liability matching? (3)
- xi) As a result of the ALM studies, various portfolios are identified, however one must look for an optimum and efficient portfolio for investment purposes. Please state the steps in identifying the optimum portfolio and the efficiency frontier. (5)
- xii) The debt assets (g-secs and bonds etc) are accounted as “held till maturity” even though about 20% of the assets are transacted every year by the fund manager. The trustees feel that under the new IFRS regime it will be better to classify debt assets as “available for sale”. Please state how the change in representation, will impact the accounts/balance sheet of the pension fund. (4)
- [60]

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